IMDM327 Final Project Report

For my final, I propose Murmuration, a single-user virtual reality experience. I've always loved birds. After growing up with screeching birdy siblings, I sometimes wonder how. My family's parrots, Tommy and Gandy, are up at the crack of dawn every day, yelling and banging for their breakfast veggies. They get jealous of each other and will hold a grudge. Last winter, I watched a movie with Gandy on my lap. Later, when I was putting them to bed, Tommy punished me, biting my finger hard. Over a year later, she is still getting reparations, ever since that night she poops on my shoulder when I'm home. They have personalities and moods, just like people. Unlike squirrels, deer, and other animals I see regularly, birds feel like individuals, each with their own routine, home, and family. It's why flocks amaze me. I'm used to talking to Gandy or seeing a sparrow or two hopping around outside, but when an incomprehensible number of crows fly overhead, I have to stop and watch. My goal for Murmuration was to capture the awe inspired by the largest of flocks on Earth.

The core architecture of the titular murmuration–the flock–is built upon a Boids simulation. Each boid, or bird in this case, relies on the positions and velocities of nearby birds. As in a standard boid simulation, birds interact through three rules: cohesion, separation, and alignment. Unlike previous projects I've worked on, optimization was one of the highest priorities to capture the sheer number of birds. I wrote the boid logic as a compute shader kernel so that many threads could run in parallel. Additionally, at the beginning of each frame, boids are sorted into cubic cells, and each boid only performs distance checks on adjacent cells. Each bird is displayed as a 32x32 texture particle using Unity's VFX Graph package. I used ShaderGraphs to create a circular ground plane and blend between it and a URP dynamic skybox texture. The

VR rig was quite simple since it was a headset-only experience, requiring nearly no modification from the default XR rig.

My biggest discovery this semester was the compute shader. I feel that it had a huge impact on my Unity skill tool belt. It also just made me feel like I was cheating the system, unlocking a secret that makes iterating through objects faster and the impossible possible. I didn't find it to be too difficult either; it really does seem like the possibilities are endless. Towards the end of the project, I began to look into VFX Graphs, which were even more intuitive due to the node-based interface. The sample scene showed awesome potential, and I will definitely continue to explore it.

Getting the opportunity to present to a crowd was great. Many people said "whoa" and "wow" when first looking at the flock, which was exactly what I intended. I spoke about my plans to add additional rules for the bird logic to hopefully increase realism, and most people thought they were worth adding, but didn't notice anything "out of the ordinary" until I pointed it out. One such thing was birds accelerating upward and downward at the same rate, which is not how they behave in real life under gravity. Some people suggested a form of interactivity, but honestly, fewer than I expected, since it can be a major aspect of immersivity. I can certainly see where they are coming from, and I had even intended to make them interactable in the project conception phase. However, I do think that interaction doesn't help the demo get closer to the goal of inspiring awe, nor does it stay true to real life. Overall, I had a great time working on the final and was glad I could talk to my classmates about it!